

GDV Report

Attack patterns from FIFA World Cup 2022 data

A practical exploratory visualization project for amateur coaches

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Short summary

This project explores simple attacking patterns from the FIFA World Cup 2022 and asks what amateur coaches can realistically take from them. I do not treat professional match data as a direct training plan. The goal is more practical: the data can make repeated patterns visible and can help coaches discuss training situations in a clearer way.

The main research question is: Which simple attacking patterns from the FIFA World Cup 2022 can amateur coaches use as ideas for training? To answer this, I analysed regular shots, goals, successful passes before shots, final third entries, entry methods and expected goals. Penalty shootouts, penalties and direct free kick shots were removed because they do not describe normal build up play.

The results show that many dangerous actions do not need a very long passing sequence. The most common shot sequences came after zero to three completed passes. At the same time, the four to six pass category had the highest goal conversion rate. For me, this was an important point: quick attacks matter, but the best situations are not only about playing fast. They are about reaching dangerous spaces with enough control to create a good shot.

1. Research question and use case

The target users are amateur coaches and football interested players with basic football knowledge. This matters because amateur teams normally have limited training time. They often cannot prepare in detail for every opponent like professional league teams do. A tournament such as the World Cup is also a useful comparison point, because teams have less time to prepare for every opponent than in a full league season. This makes the question relevant for amateur football: should a team focus on longer possession phases, faster counter attacks, or a mix of both?

My personal motivation also came from my own football context. In amateur football, many coaches and players watch professional teams and try to copy their style. Sometimes this works, but sometimes it does not fit the level, the training time or the available players. My own team often tries to build attacks with many passes and more possession. This can look controlled, but it can also become slow and make it harder to create clear chances. That is why I wanted to check how often shots and goals at World Cup level actually come from short, medium or long build ups.

The analysis is therefore not meant to say that one style is always correct. It is meant to support coaching discussion. Examples are the first forward pass after winning the ball, support runs after entering the final third and quick occupation of shooting areas. These are simple ideas that can be trained without needing a professional tactical setup.

2. Data and preparation

The project uses StatsBomb Open Data from the FIFA World Cup 2022. The data is event data. This means that the dataset contains actions on the ball, such as passes, carries and shots, together with pitch coordinates and additional information such as expected goals. Expected goals, or xG, estimate the chance that a shot becomes a goal based on similar shots. A higher xG value means better chance quality, but it is still an estimate and not a guarantee.

From the raw data, I created three working tables. The first table contains regular shot sequences. In total, the final analysis includes 1145 regular shots and 123 goals. For each shot, I counted how many completed passes happened earlier in the same possession. The second table contains 7456 final third entries. An entry is counted when the ball is moved into the final third, either by pass, carry or by a possession that already starts high. The third table contains team level summaries, for example shot rates after entries and average completed passes before goals.

The most important data decision is the treatment of special shot situations. Penalty shootouts, penalties and direct free kick shots are excluded from the main analysis. They can create shots or goals with no normal build up and would make the short pass categories look stronger than they actually are in open attacking play. Corners and throw ins remain in the data because they can still continue as regular attacking situations.

The main limitation is that event data only shows actions on the ball. It does not show all off ball runs, pressing structure or the exact instructions from the coach. For that reason, the results are interpreted as descriptive patterns and not as proof that one attacking method is always better.

3. Design approach and GDV choices

The report follows a simple visual story. First, it shows how often different build up categories lead to shots and goals. Then it separates volume from efficiency. After that, it moves to final third entries, entry methods, start zones and team directness. The last figure is a Spain case study that connects the general patterns to a concrete team example.

For category comparisons, I used bar charts because values on a common baseline are easy to compare. This fits the idea from graphical perception research that length and position are usually easier to read than area or angle. For team comparisons, I used a lollipop style plot because it keeps many team names readable without making the figure too heavy. For the Spain example, I used a pitch plot because the location on the field is part of the message.

I avoided visualizations that looked impressive but did not make the coaching message clearer. Earlier heatmap ideas were interesting, but they were harder to explain to the target audience. The evaluation also confirmed that simple figures were understood faster, while the Spain plot needed a stronger caption because it contains several visual elements at once.

4. Results

Figure 1 shows the volume of regular shots and goals by build up category. The most common category is zero to three completed passes, with 506 regular shots and 43 goals. This supports the idea that short attacking sequences are common. It does not mean that every team should only play direct football, but it shows that quick progression to goal is an important part of chance creation.

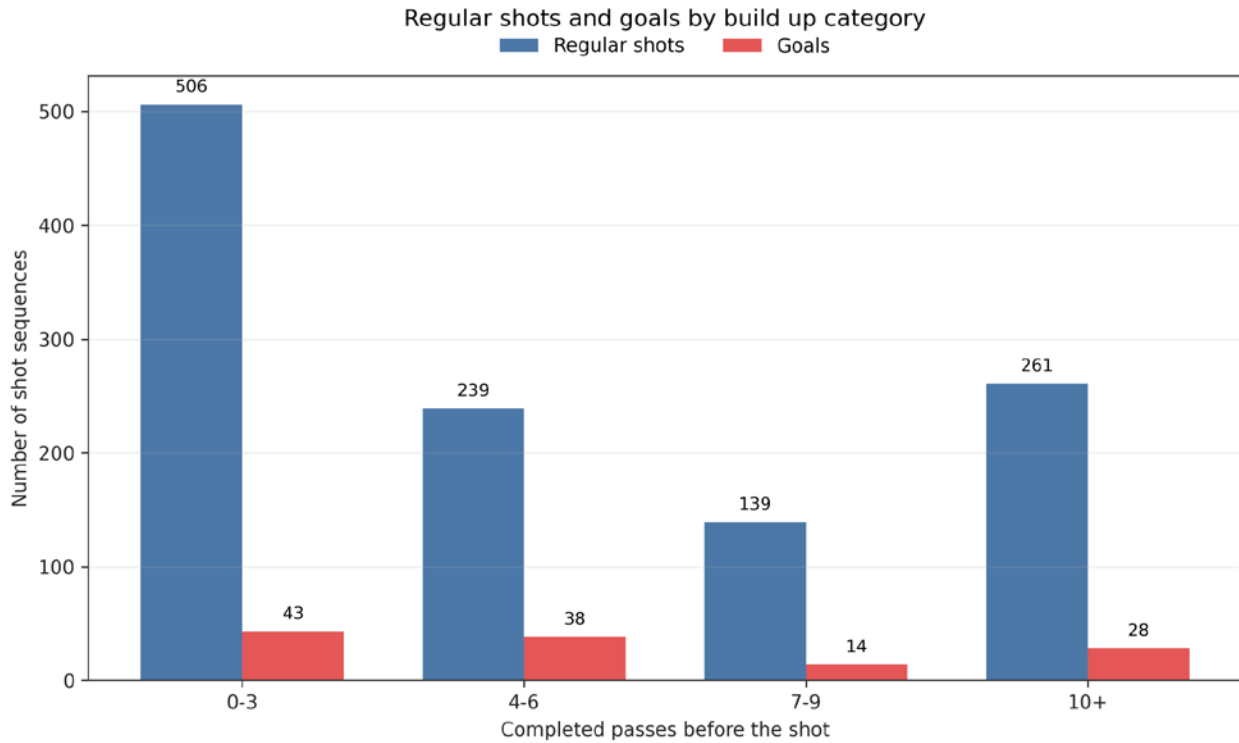


Figure 1. Regular shots and goals by build up category. The figure shows the number of regular shot sequences and goals after different numbers of completed passes.

Figure 2 separates frequency from efficiency. The four to six pass category has the highest goal conversion rate at 15.9 percent and also the highest average xG per shot in this comparison. This is useful because the most frequent pattern is not automatically the most efficient one. A practical training idea is therefore not only to play quickly, but to play quickly enough while still creating a controlled shooting position.

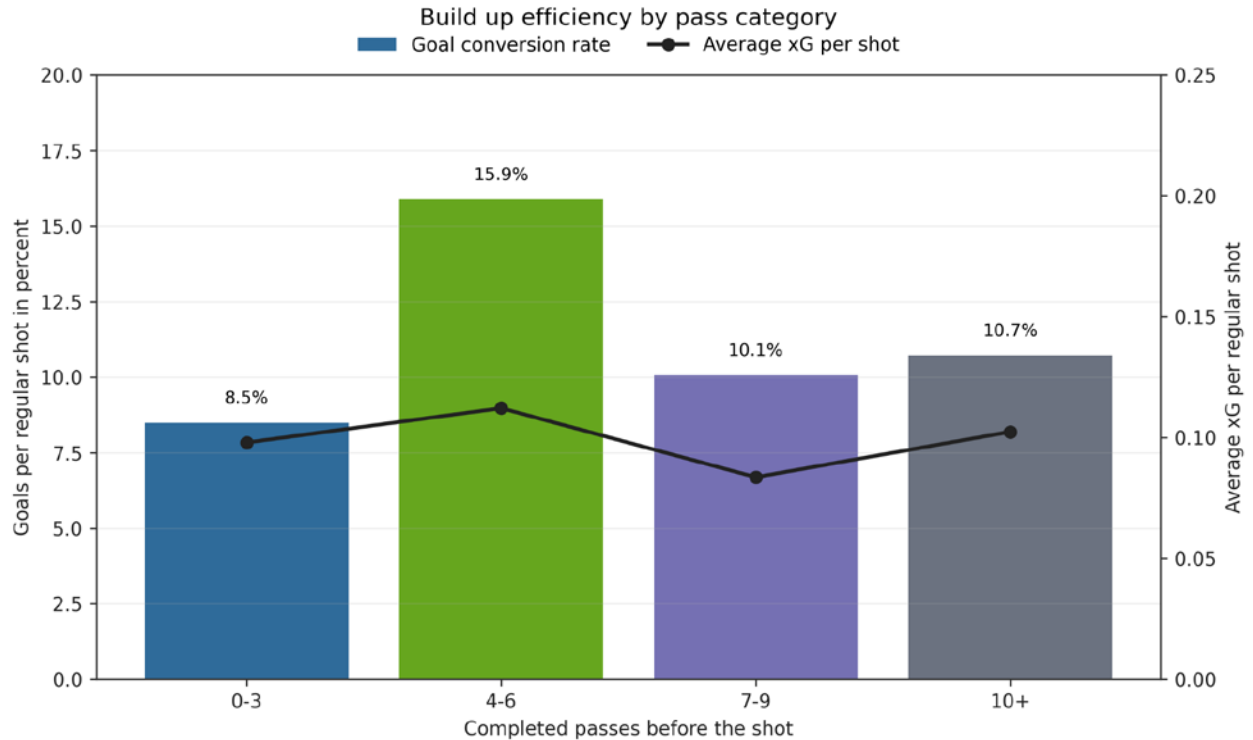


Figure 2. Build up efficiency by pass category. Bars show goal conversion rate. The line shows average xG per regular shot. The two axes should not be compared directly.

Figure 3 looks at teams that turned final third entries into shots. France, Germany and Belgium are among the teams with high shot rates after entries in this tournament data. This does not rank team quality. It shows how often a team created a shot after moving the ball into a dangerous area.

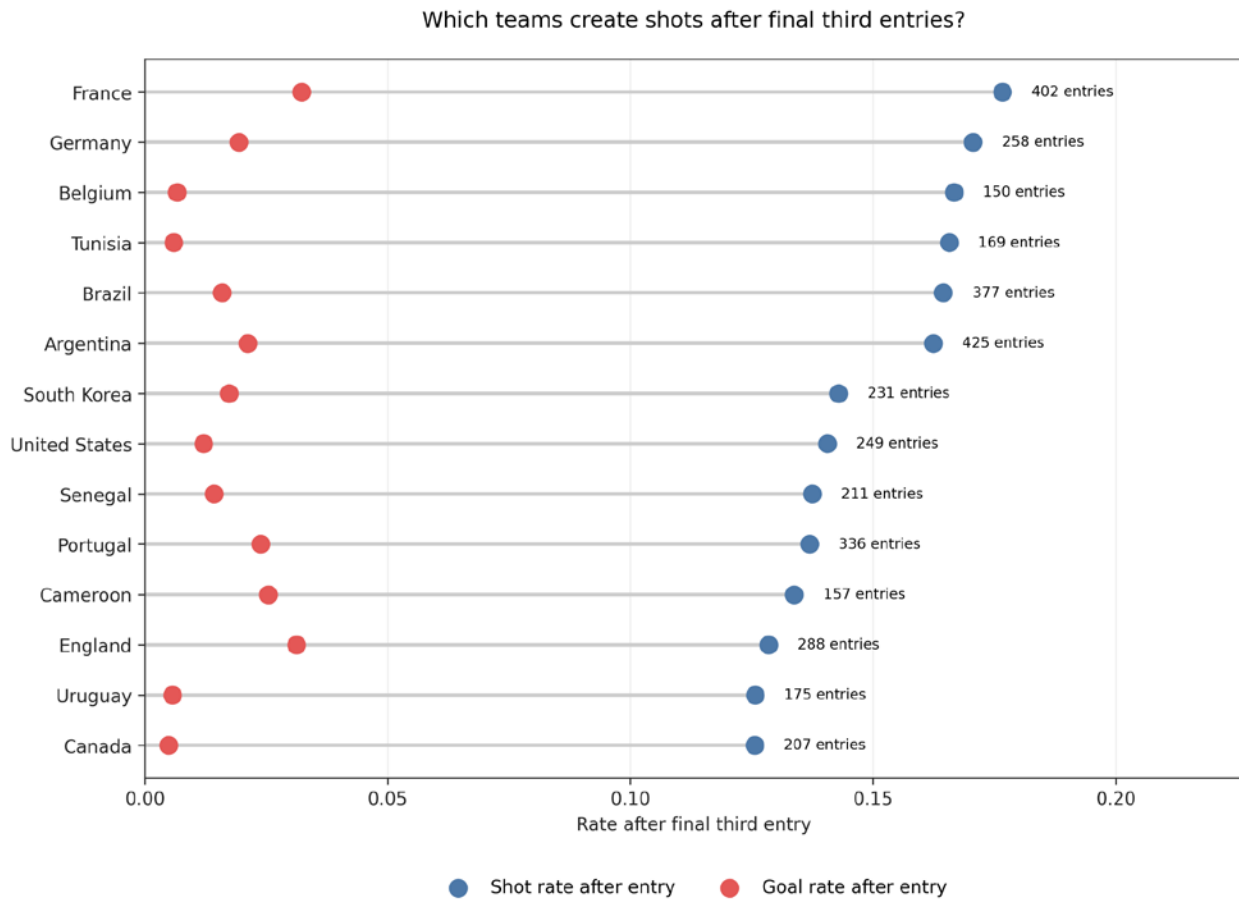


Figure 3. Shot and goal rates after final third entries. The plot connects progression into dangerous areas with the outcome after the entry.

Figure 4 compares entry methods. Carries had the highest shot rate after entry at 19.3 percent. Long passes also had a strong shot rate at 15.5 percent. I interpret this carefully. It does not prove that carries or long passes are always better, but it gives useful training scenarios: move the ball forward, support the player quickly and create a shot option before the opponent is fully organized.

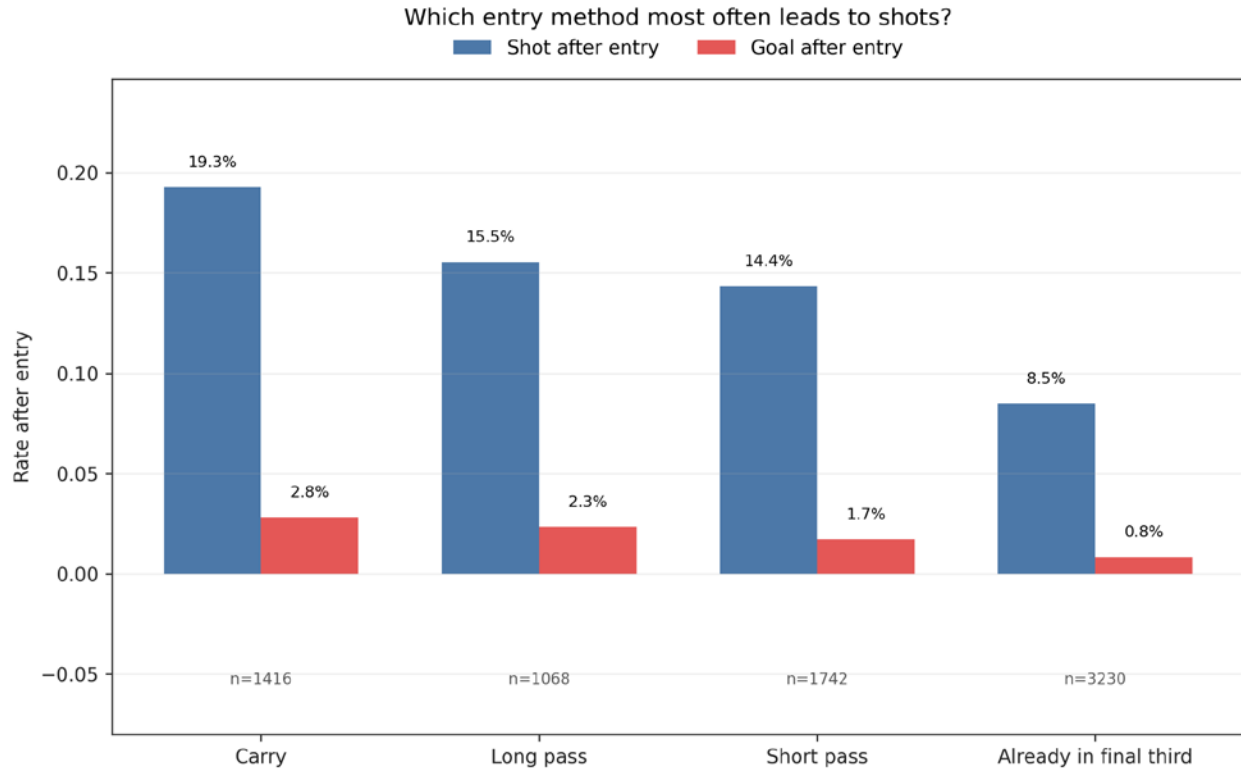


Figure 4. Entry method and outcome. The figure compares carries, long passes, short passes and possessions that already start in the final third.

Figure 5 simplifies the start zones of goal attacks. This is less detailed than a heatmap, but it is easier for the target audience to read. Because the plot only uses goals, some categories contain fewer examples. I therefore use it as a simple overview and not as a strong statistical comparison.

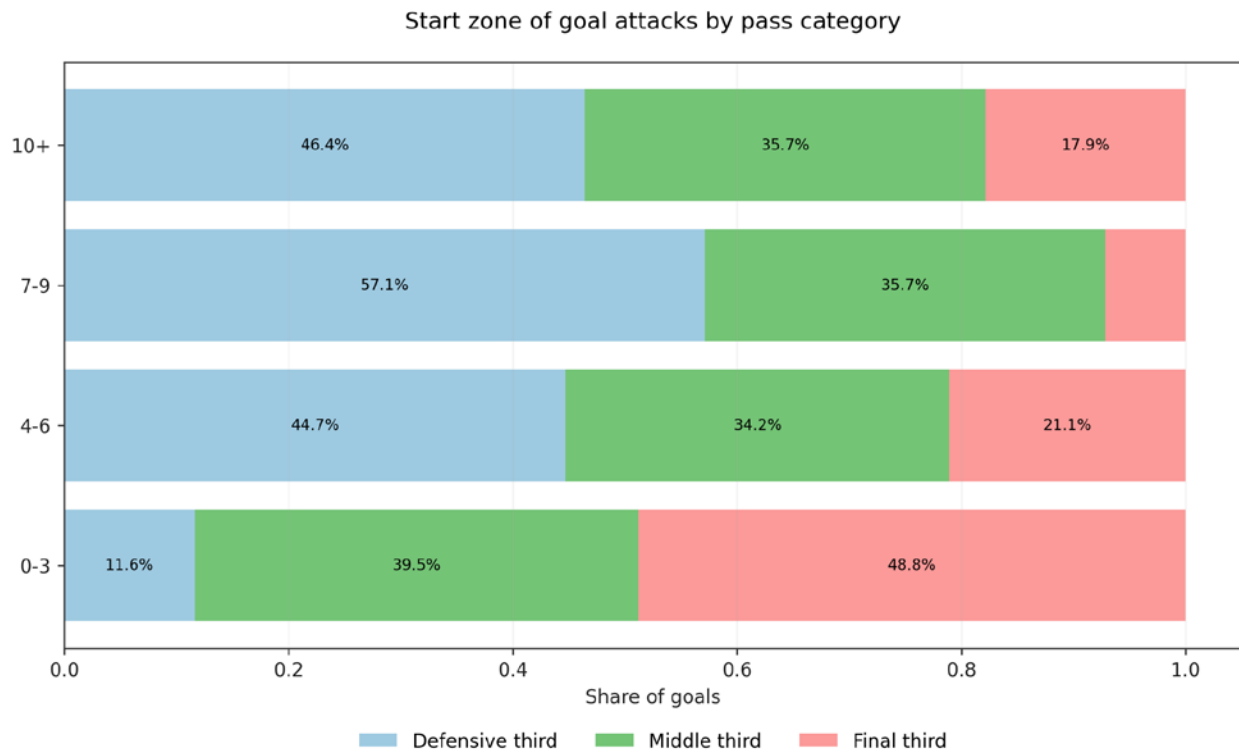


Figure 5. Start zone of goal attacks by pass category. The simplified zones make the origin of goal attacks easier to explain.

Figure 6 compares team directness. Only teams with at least two analysed goals are shown. Spain is a clear outlier with a much higher average number of completed passes before goals. This fits Spain as a possession based example, but the figure is still only a style comparison. A team with fewer passes before goals is not automatically better than a team with longer build up play.

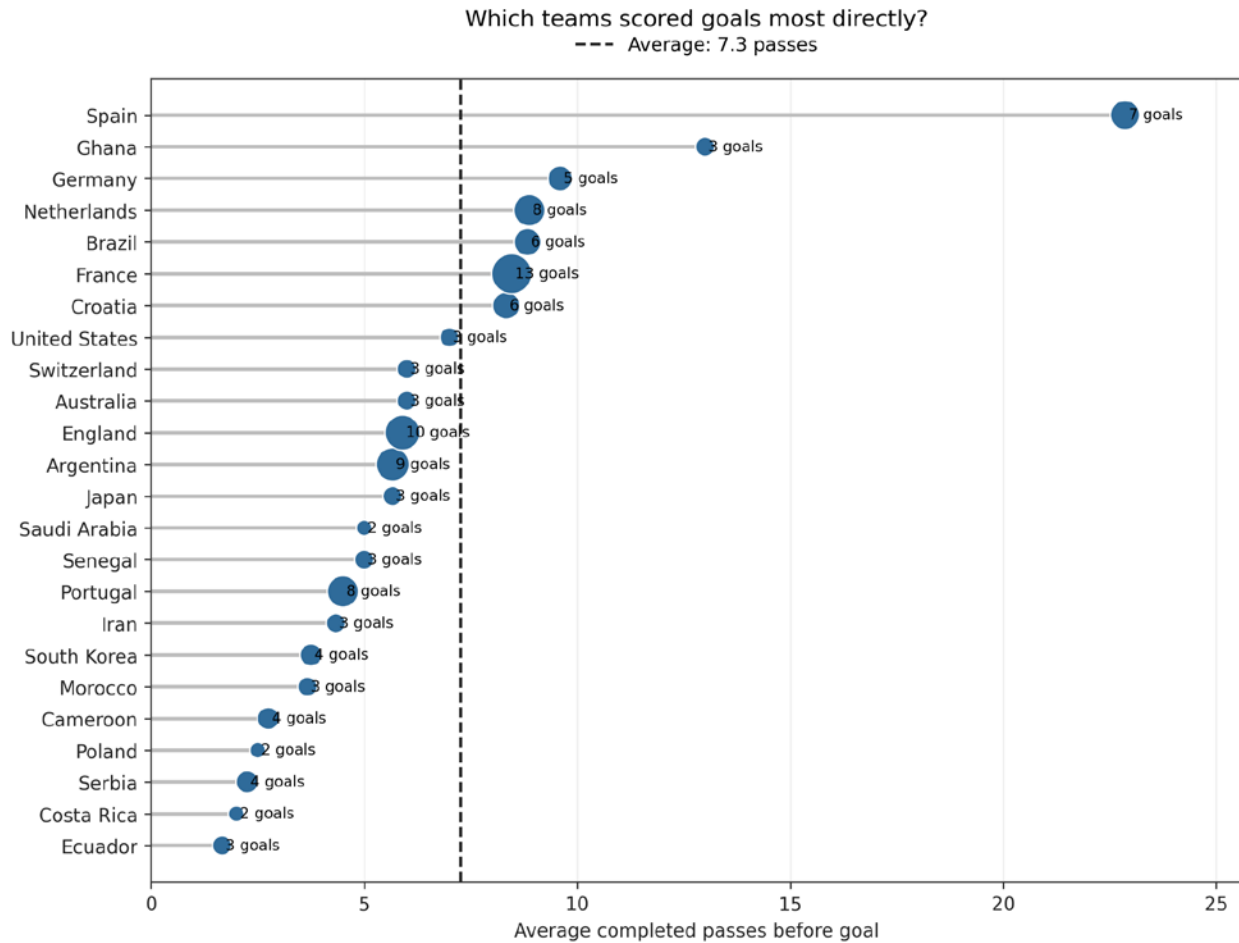


Figure 6. Team directness as a style comparison. The lollipop plot shows the average number of completed passes before goals.

Figure 7 uses Spain as a concrete case study. Spain was useful for this report because it connects possession based play with chance creation. The plot shows entries into the final third that led to shots. The darker green zones show more summed xG after entries from that lane. For an amateur coach, the training idea is not to copy Spain exactly, but to train repeated entries, support around the ball and quick occupation of central and half space shooting areas.

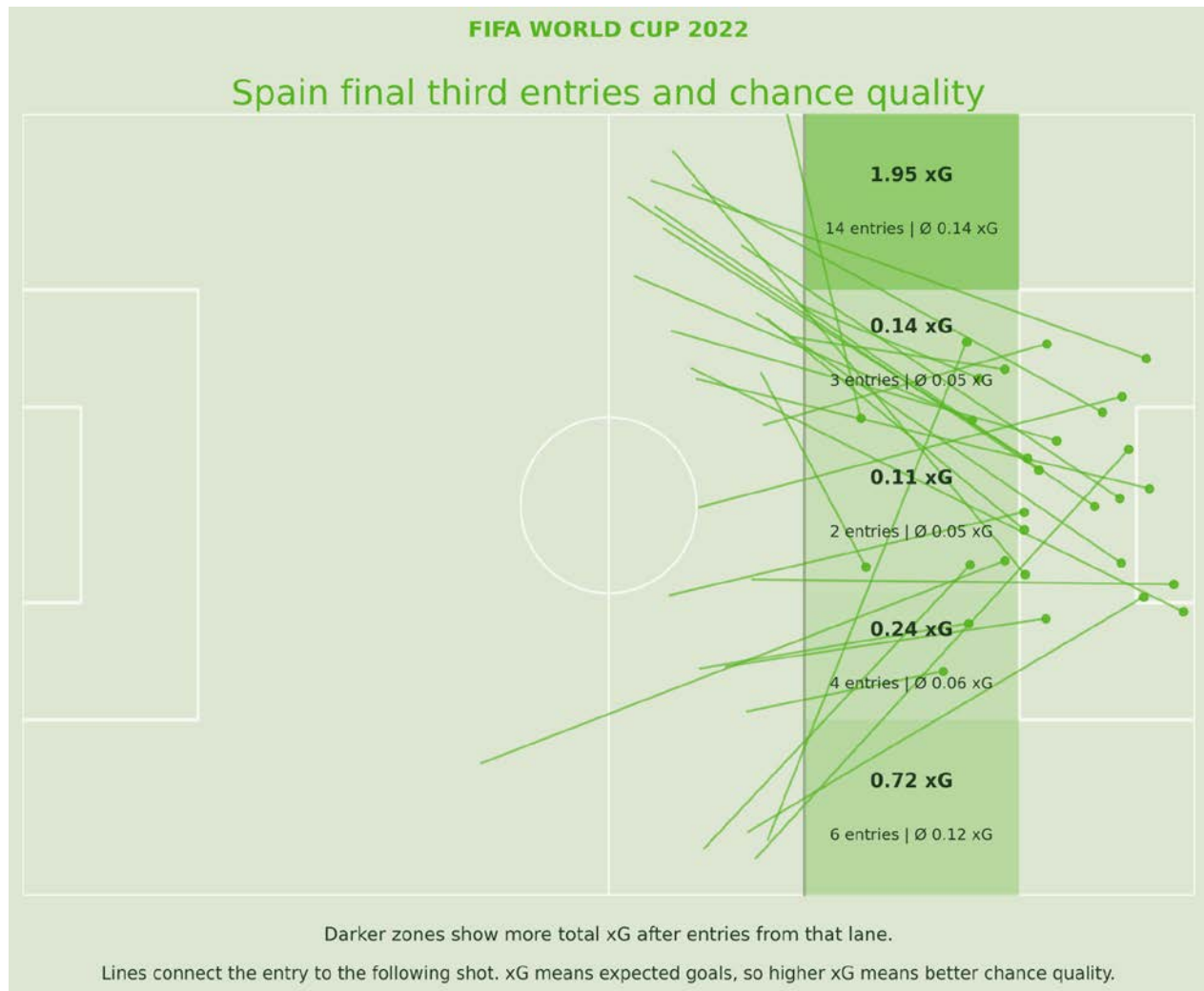


Figure 7. Spain final third entries and chance quality. Lines connect the entry to the following shot. Darker green zones show more total xG after entries from that lane.

5. Evaluation and iteration

I conducted a short formative evaluation with two teammates who play football. They looked at the final figures and answered simple questions about pass categories, conversion rate, entries, entry methods, team directness and the Spain plot. The goal was not to run a statistical study. The goal was to check whether football players could understand the figures and where the wording needed to be clearer.

The simple comparison charts were understood fastest. One participant said that the first plot was clear because the highest bars are visible immediately. The second participant found the goal conversion plot easy because it focused on one main comparison. The more complex figures needed more explanation. The word entry was not immediately clear, and the Spain plot was interesting but visually fuller than the other plots.

Based on this feedback, I improved the wording around the figures. I explained entry more simply, added a direct xG explanation in the Spain figure and avoided wording that makes the plots sound like fixed tactical rules. This made the report more honest and easier to read.

6. Limitations and conclusion for training

The results should be interpreted carefully. Event data shows actions on the ball, but not all movement around the ball. It also cannot show the coach instructions behind an attack. Opponent strength, game state, match phase and player quality can influence every result.

The main training conclusion is not that teams must always play direct. The better conclusion is that teams should have clear ideas for moving the ball into dangerous areas and creating a shot soon after. This can be trained through exercises around the first forward pass after winning possession, support after entering the final third and quick occupation of shooting areas.

From a GDV perspective, the project also shows that simple and well explained visualizations are often more useful than complex graphics that look impressive but are harder to read. The best figures in this project are the ones where the reader can quickly understand both the result and the limitation.

References

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